

Resource competition

Learning objectives

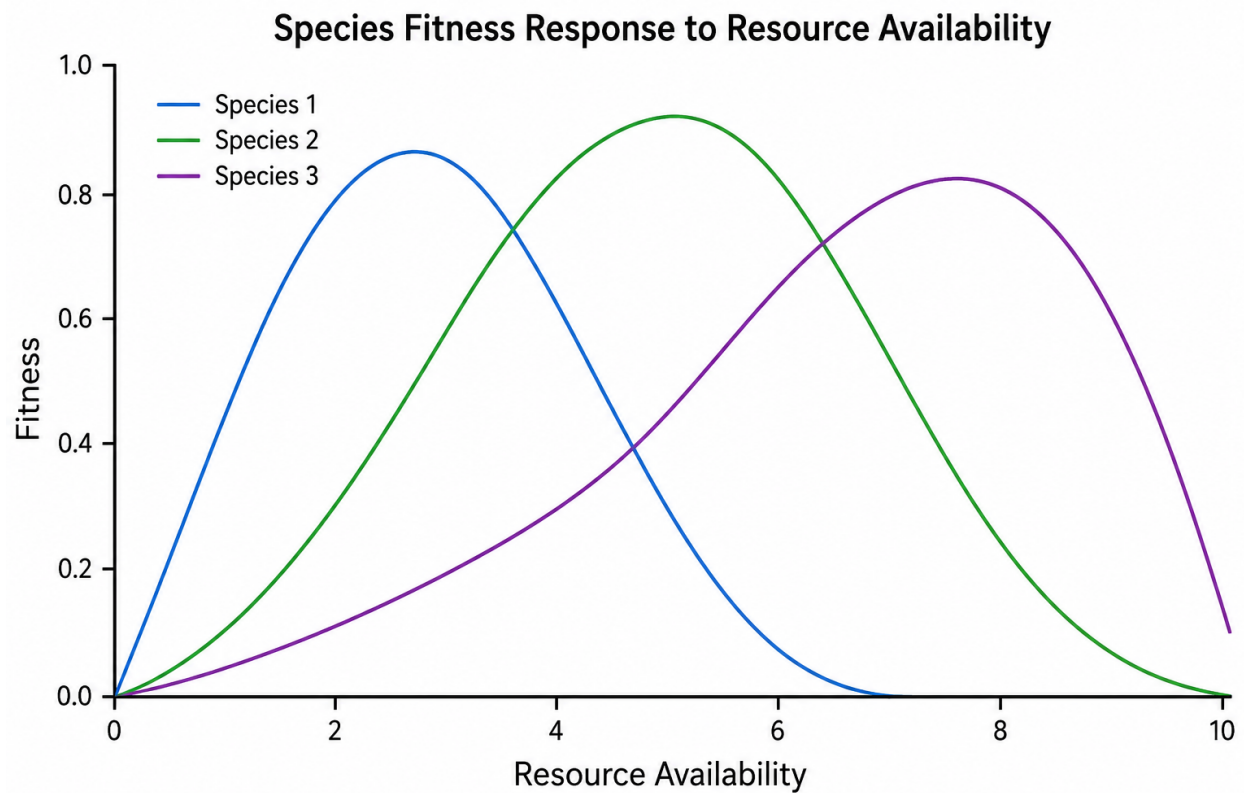
1. Understand how resource availability affects coexistence
2. Know the strengths and limitations of Tilman's R^* theory and MacArthur's consumer resource model

last lecture: phenomenological competition model, only effects, not actual resource levels

Resource competition and diversity

talked about mechanisms of diversity maintenance, but didn't model any mechanisms or processes

- optimal resource or level varies by species
- environmental heterogeneity creates resource variability



- niche overlap *color in overlap in distribution*

Neutral theory

because we've talked about niche theory, I'll also mention neutral theory

- assumes all species equal in response to resources
- diversity maintained from speciation and stochastic extinction

Modeling resource competition

Tilman's R^* theory

- R^* = amount of resource a species needs to maintain equilibrium
- if R^* of species 1 $<$ R^* of species 2 for same resource, species 2 excluded
- 2 resources - stable coexistence if
 - limited by different resources
 - consume more of that limited resource than other species
- assumptions: constant resources, equilibrium

related to intraspecific competition $>$ interspecific competition - species self-limit

MacArthur's consumer-resource model

- model consumer and resource density *traditionally for predator-prey dynamics*
- attack rates = consumption
- multiple resources and consumers
- assumptions: consumer abundance $\leq$ resource abundance